

EXCEPTION – 06.08.2026
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SIMATS Online Programming Lab
JAVA PROGRAMMING

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QUESTIONS

Q1
Handling ArithmeticException
10 marks

Q2
Handling
ArrayIndexOutOfBoundsException
10 marks

Q3
Handling
NumberFormatException
10 marks

Q4
Handling Multiple Exceptions
10 marks

4/4 answered

Q1 Handling ArithmeticException

10 marks

Write a Java program to divide two integers entered by the user. If the denominator is zero, handle the exception using a try-catch block and display an appropriate error message.

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         try {
7             System.out.print(sc.nextInt() / sc.nextInt());
8         } catch (ArithmeticException e) {
9             System.out.print("Error: Division by zero");
10        }
11    }
12 }
```

Run Save

Input

12
0

Output

Error: Division by zero

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4/4 answered

Q2 Handling ArrayIndexOutOfBoundsException

10 marks

Write a Java program to create an integer array of five elements. Ask the user to enter an index and display the corresponding element. If the index is outside the valid range, handle the exception and display an error message.

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         int[] arr = {10, 20, 30, 40, 50};
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter an index: ");
9         int index = sc.nextInt();
10
11        try {
12            System.out.println("Element at index " + index + " is: " + arr[index]);
13        } catch (ArrayIndexOutOfBoundsException e) {
14            System.out.println("Error: Index out of bounds! Please enter a valid index.");
15        }
16
17        sc.close();
18    }
19 }
```

Run Save

Input

10

Output

Compilation Error:

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4/4 answered

Q3 Handling NumberFormatException 10 marks

Write a Java program to accept a number as a string and convert it into an integer using Integer.parseInt(). If the input is not a valid integer, handle the exception and display an appropriate message.

Your Code

```
1 import java.util.Scanner;  
2  
3 public class NumberFormatHandler {  
4     public static void main(String[] args) {  
5         Scanner scanner = new Scanner(System.in);  
6  
7         System.out.print("Enter a string to convert to an integer: ");  
8         String input = scanner.nextLine();  
9  
10        try {  
11            int number = Integer.parseInt(input);  
12            System.out.println("Successfully converted: " + number);  
13        } catch (NumberFormatException e) {  
14            System.out.println("Error: The input is not a valid integer");  
15        } finally {  
16            scanner.close();  
17        }  
18    }  
19 }
```

Input

"123"

Output

Enter a string to convert to an integer; Error: The input is not a valid integer.

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Q4
Handling Multiple Exceptions
10 marks

4/4 answered

Q4 Handling Multiple Exceptions 10 marks

Write a Java program that performs the following operations:
Accept two integers from the user.
Divide the first number by the second.
Store the result in an array.
Handle both ArithmeticException and ArrayIndexOutOfBoundsException using multiple catch blocks.

Your Code

```
1 import java.util.Scanner;  
2  
3 public class MultipleExceptionHandling {  
4     public static void main(String[] args) {  
5         Scanner scanner = new Scanner(System.in);  
6  
7         try {  
8             System.out.print("Enter first number: ");  
9             int num1 = scanner.nextInt();  
10  
11            System.out.print("Enter second number: ");  
12            int num2 = scanner.nextInt();  
13  
14            int result = num1 / num2;  
15            int[] resultArray = new int[1];  
16            resultArray[0] = result;  
17  
18            System.out.println("Attempting to access index 1....");  
19            System.out.println(resultArray[1]);  
20  
21        } catch (ArithmeticException e) {  
22            System.out.println("Error: ArithmeticException caught! (Caused by: " + e.getMessage());  
23        } catch (ArrayIndexOutOfBoundsException e) {  
24            System.out.println("Error: ArrayIndexOutOfBoundsException caught! (Caused by: " + e.getMessage());  
25        } catch (Exception e) {  
26            System.out.println("An unexpected error occurred: " + e.getMessage());  
27        } finally {  
28            scanner.close();  
29            System.out.println("Program execution completed.");  
30        }  
31    }  
32 }
```

Input

10
3

Output